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Branding is what people say about you when you are not in the room.

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-Jeff Bezos



The four key takeaways from this article

1. Semiconductors are at the heart of the new digital economy, which is destined to keep growing for many years to come.
2. There is a shift from fixed-function hardware to programmable platforms. Even the networks are moving towards software and becoming more programmable and flexible. Developers who own a network infrastructure can more rapidly improve and optimise their network to make it more reliable and secure, and differentiate it from the competitors'.
3. Silicon will power the next generation of infrastructure innovation that will simplify the work of developers, amplify their capabilities, and improve their efficiency.
4. The new networks and programmable platforms are also embracing open tool chains, open software frameworks, and open APIs.

at the very highest performance from a software infrastructure. They felt it would run slower and consume more power, but it turns out that they were wrong.

Rakatan's recent study measured how quickly users could download data on different 5G networks. In Japan, right in downtown Tokyo at the busiest crosswalk in the world, Shibuya crossing, they found that Rakatan's VRAN network has 40% faster speed than any other major download speed on a different Rakatan's VRAN network. With a virtualised network running on software, no one would have predicted that the biggest mobile networks in the world would have become software-defined.

The last piece of the network RAN is moving towards software, as it's driven by the need for flexibility, programmability, versatility, and the need to make these systems more reliable and more secure. Only those who operate networks at a huge scale understand how to do so, and only those who build and operate that infrastructure can quickly introduce new capabilities and differentiate from their other customers.

The modification from fixed-function hardware to software doesn't stop at the network but is also implemented in factory automation systems, where a plethora of small embedded microcontrollers are being replaced by sensors and actuators that are centrally controlled by an open computing platform. Inference applications run on top of Kuber-

netes. Similarly, in warehouses, real-time in retail stores, and even in cars, computing is being centralised to create software-defined cars.

Likewise, 5G is another interesting example, as it is intertwined across this entire landscape. 5G and 6G in the future are promising to unlock an industry of new applications, smart cities, new factory automation and pretty much autonomous everything. Because of its high bandwidth, low latency, and deterministic and strong security model, 5G is really an ideal platform for landing these new applications.

Increasing relevance of silicon with demand for software

All software runs on hardware. The silicon actually defines the performance, the power, and the ability to write programs and then compile them down to a target. Hence, contrary to the assumption that silicon is becoming less relevant, the fact remains that, with increased performance and scale that's being driven by technologies like 5G and edge inference, silicon will become more relevant over time.

We need silicon that's more performant than ever. Various acceleration technologies for security storage inference are going to be critical, and this is where we are going to continue to invest. **EFY**

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